Design and Development of a Cost-Effective Infant Incubator for Underprivileged Areas

Final Year Project Portfolio

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Abstract

High infant mortality in underprivileged areas is strongly linked to the lack of affordable neonatal incubators. This project focuses on designing and developing a low-cost, portable, and energy-efficient infant incubator. The prototype ensures safe temperature (35–37°C) and humidity (40–60%) control using Arduino, sensors, and simple materials. Test results show that the incubator is cost-effective, lightweight, and functional compared to traditional incubators. The project contributes to Sustainable Development Goal (SDG) 3.2 by reducing preventable neonatal deaths, empowering communities, and improving healthcare access.

Introduction

Premature and low-birth-weight infants face major survival challenges due to limited access to incubators in resource-constrained areas. Conventional incubators are costly, non-portable, and difficult to maintain. This project proposes a sustainable biomedical engineering solution to bridge this gap. By integrating affordable electronics and materials, the incubator supports neonatal care in rural and underserved hospitals. This aligns with my career goal of contributing innovative biomedical solutions globally.

Objectives

1. To design a cost-effective infant incubator for low-resource settings.

2. To ensure portability and easy maintenance.

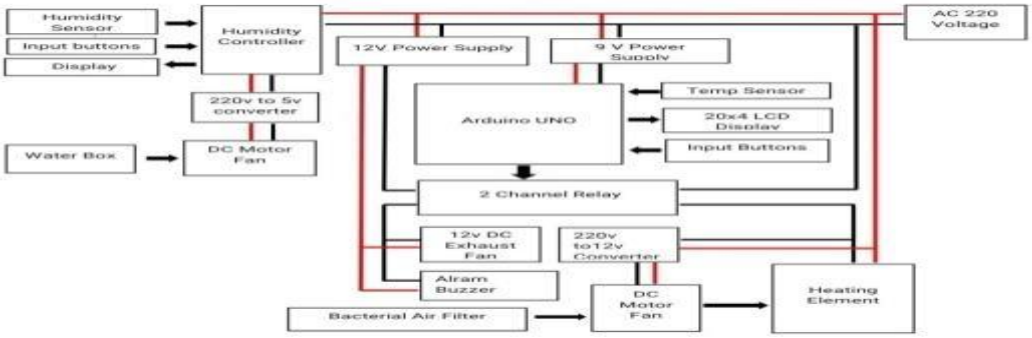
3. To provide safe monitoring of temperature and humidity.

4. To contribute to neonatal survival and healthcare equity.

Methodology / Design Approach

The design process included:

• Literature review on neonatal care and incubator designs.
• Block diagram and CAD design of the incubator.
• Selection of low-cost components (Arduino Uno, DHT11 sensor, ceramic heating element, LCD display, humidity controller).
• Circuit design, breadboard testing, and Vero-board integration.
• Prototype assembly with acrylic sheets and insulated materials.
• Testing for temperature and humidity stability.



Prototype & Results

Key prototype features:

• Temperature maintained at 35–37°C.

• Humidity maintained at 40–60%.

• Lightweight and portable (<15kg).

• Operates on 12V DC supply with low power consumption.

Figure 3.2: Arduino UNO

Figure 3.3: 20× 4 LCD

Figure 3.4: 12-volt power supply

Figure 3.5: 12 LM2596

Figure 3.6: 5v Dual Channel Relay

Figure 3.7: Ceramic Heating Element

Figure 3.8: Potentiometer 10k

Figure 3.9: Humidity Controller

Figure 3.10: DC 12v fan

Figure 3.11: DHT11 Sensor

DESIGN AND DEVELOPMENT

4.1 Circuit Design

4.1.1 Arduino connection with LCD

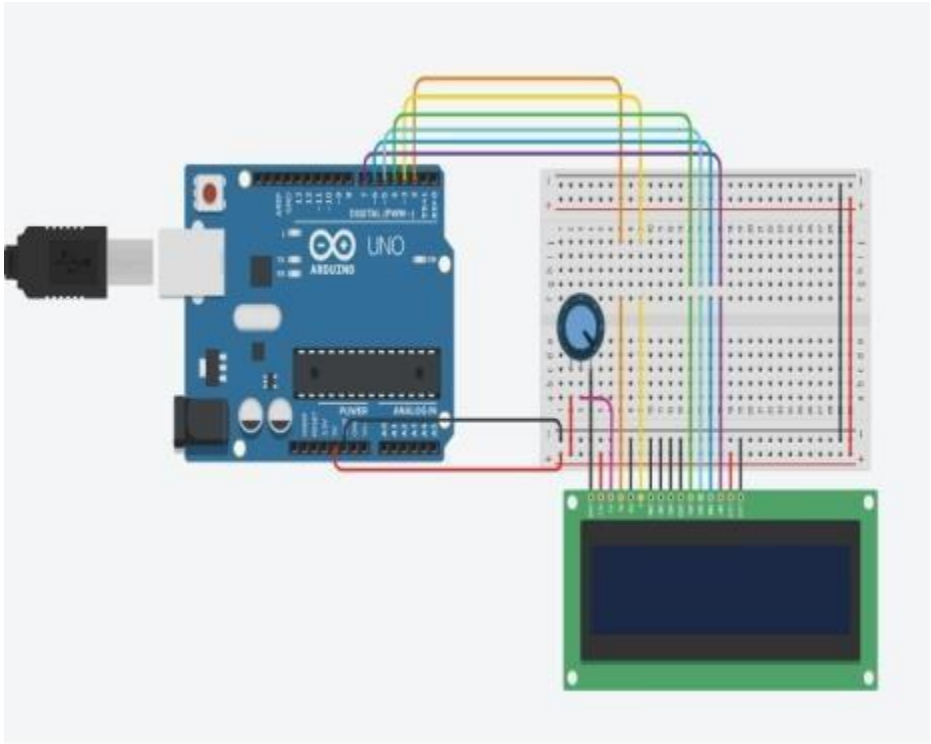
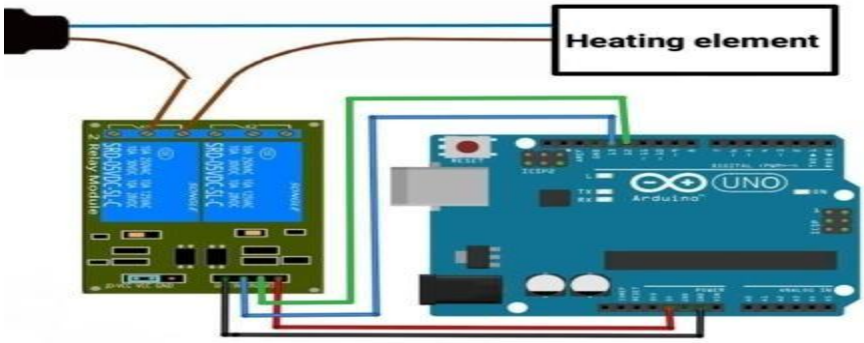


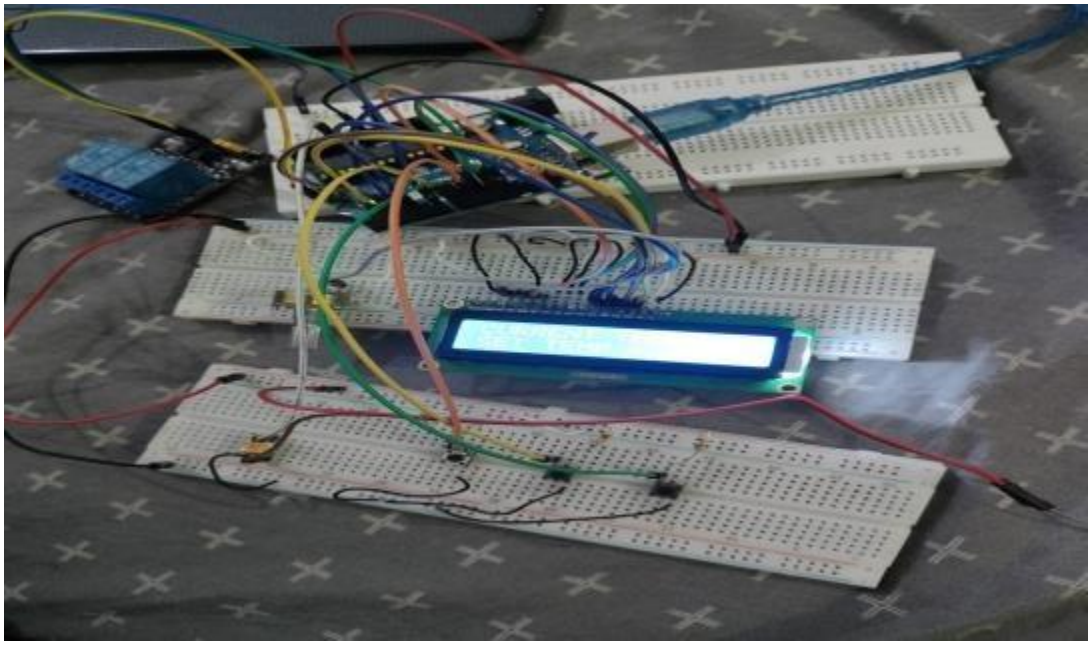
Figure 4.1: LCD connection with Arduino UNO



4.1.2 Arduino connection with Relay

Figure 4.2: 2 Channel Relay connection with A

4.1.4 Breadboard connection of hardware



Parameter	Traditional Incubator	Our Prototype
Cost	High ($5,000–$15,000)	Low (<$300)
Portability	Bulky, immovable	Lightweight, portable
Power Consumption	High	Low
Maintenance	Complex, costly	Simple, low-cost

Impact

Our designed incubator addresses healthcare gaps by offering an affordable neonatal care solution. It directly supports SDG 3 (Good Health and Well-being), specifically target 3.2: ending preventable deaths of newborns and children under 5. By lowering costs and increasing accessibility, it relieves financial pressure on families and healthcare systems while saving lives.

Skills Gained

• Research and literature review in biomedical engineering.

• Design and prototyping using Arduino and sensors.

• CAD design and hardware integration.

• Problem-solving and teamwork in a multidisciplinary environment.

Future Scope

Future improvements include:
• Integration of IoT for remote monitoring.
• Use of AI for predictive health alerts.
• Solar-powered operation for rural hospitals.
• Advanced sensors for oxygen monitoring and infection prevention.

Conclusion

The project demonstrates how biomedical engineering can provide affordable, practical, and impactful solutions for global health challenges. This experience has motivated me to pursue advanced studies in biomedical engineering, where I aim to expand my expertise, work in diverse research teams, and contribute to global innovations in healthcare technology.

5.1 Portable Infant Incubator

Infant Incubator Testing when Start

Fig: Compare Temp value with Temp meter at start level

Compare temp & humidity values with Temp meter

Figure 5.4: compare temp & Humidity values with meter

References

[1] WHO. Newborn Mortality Statistics, 2023.

[2] Tarnier, S. Historical Development of Infant Incubators, 1880.

[3] Arduino.cc, Arduino Uno Technical Documentation.

[4] DHT11 Sensor Datasheet.

[5] UNICEF, Improving Neonatal Care in Low-Resource Settings, 2022.